

Short-term photovoltaic solar power forecasting using a hybrid Wavelet-PSO-SVM model based on SCADA and Meteorological information

Abinet Tesfaye Eseye^{a, b, *}, Jianhua Zhang^a, Dehua Zheng^b

^a School of Electrical and Electronic Engineering, North China Electric Power University, Changping District, Beijing, 102206, China

^b Microgrid Platform R&D Department, Goldwind Science and Etechwin Electric Co., Ltd, BDA, Beijing, 100176, China

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ABSTRACT

Photovoltaic (PV) solar power generation is always associated with uncertainties due to solar irradiance and other weather parameters intermittency. This creates a huge barrier in integrating solar power into the grid and biases power industries against deploying PV systems. Thus accurate short-term forecasts are important to efficiently integrate PV systems into the grid. This paper proposes a hybrid forecasting model combining wavelet transform, particle swarm optimization and support vector machine (Hybrid WT-PSO-SVM) for short-term (one-day-ahead) generation power forecasting of a real microgrid PV system. The model is developed by incorporating the interactions of the PV system Supervisory Control and Data Acquisition (SCADA) actual power record with Numerical Weather Prediction (NWP) meteorological data for one year with a time-step of 1 h. In the proposed model, the wavelet is employed to have a considerable impact on ill-behaved meteorological and SCADA data, and SVM techniques map the NWP meteorological variables and SCADA solar power nonlinear relationship in a better way. The PSO is used to optimize the parameters of the SVM to achieve a higher forecasting accuracy. The forecasting accuracy of the proposed model has been compared with other seven forecasting strategies and reveals outperformed performance with respect to forecasting accuracy improvement.

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1. Introduction

Renewable energy sources particularly solar and wind have gained more concern worldwide in recent years as they are easy accessible (available everywhere), zero-emission (clean energy and environmental friendly) and simple (less complex structure as conventional energy sources). This has greatly accelerated the installation of photovoltaic (PV) solar systems globally. However, a PV system power output is stochastic and intermittent in nature, as it relies on solar radiation and other weather variables. This stochastic behavior creates several problems to safely operate and dispatch the power network [1]. PV solar systems' output power forecasting is therefore a challenging task as it highly relies on external circumstances such as solar radiation and other weather

parameters, which are entirely intermittent and volatile in nature. An accurate PV power forecasting system is useful for the dispatching center of power networks and energy trading companies to make accurate decisions on critical issues such as alternate adjustments for conventional power sources, scheduling arrangements, storage requirements, and overall planning.

Thus, accurate short-term output power forecast of PV systems in large power grid or microgrids plays a key role for efficient, economic, stable and sustainable operation of the power supply. Several techniques have been developed recently to forecast solar radiation and PV system output power. Existing PV solar power forecasting methods can be categorized as physical and statistical methods based on the forecasting models they used. The physical methods estimate the PV power production based on the meteorological information at the PV cell (solar irradiation, temperature, cloud cover, humidity, pressure, wind speed and rain fall, etc.), which is more complex but more accurate and practical.

Whereas, the statistical methods estimate the PV power production based on historical records of PV systems' power output data, which is more commonly used. The statistical technique is

* Corresponding author. School of Electrical and Electronic Engineering, North China Electric Power University, Changping District, Beijing, 102206, China.

E-mail addresses: abinet.tesfaye@gmail.com (A.T. Eseye), jhzhang001@163.com (J. Zhang), zhengdehua01@hotmail.com (D. Zheng).

further classified into direct and indirect forecasting techniques. The indirect technique is to estimate the PV system output power based on the forecasted value of solar irradiation and other meteorological variables at the PV system. The direct technique is to forecast the power output directly based on historical power output data as input, which is commonly used. Moreover, according to the mathematical model they employed, the direct technique can be classified into time series forecasting methods [2], the regression model forecasting method [3], neural network forecasting method [4], Markov Chain forecasting method [5] and etc.

Among these techniques, the artificial intelligence techniques, particularly the neural network and the probability forecasting algorithms, are broadly used recently. References [6]–[7] have established back propagation (BP) neural network model to predict the PV power output by investigating the impact of solar irradiation, temperature, cloud cover and other meteorological variables on the PV output power. Ref. [8] employs dynamic Bayesian network model to predict the probability distribution of the PV system short-term power output, after examining interactions among the different factors influencing the PV system. Ref. [9] employs support vector machine (SVM) algorithm to establish a forecasting model, by investigating the impact of solar irradiation and temperature on the PV solar power system, studies and compares forecasting accuracy of two different types of weather circumstances in Italy. Ref. [10] employs random forest and quantile regression algorithms for modeling the PV power forecasting system, considering historical PV power generation data and meteorological measurement data as inputs to obtain the numerical forecast and the probability forecast of PV power system.

Ref. [11] presents a least-square support vector machine (LS-SVM)-based model for short-term PV solar power forecast utilizing two-dimensional (2D) transmissivity data and other meteorological variables, comprising cloud cover, relative humidity, and wind speed. The model outperforms autoregressive (AR) and radial basis function neural network (RBFNN) based models regarding forecasting accuracy. Ref. [12] presents a weather-based hybrid technique for one-day ahead hourly prediction of PV power output. It consists of classification, training, and forecasting stages. It used self-organizing map (SOM) and learning vector quantization (LVQ) networks for classifying the collected data and support vector regression (SVR) for training the input-output data sets. A multitime-scale data-driven short-term PV power prediction model is proposed in Ref. [13] to enhance the forecast accuracy. The model combines both spatial and temporal correlations among neighboring solar power plants, and is revealed to have enhanced performance compared to the persistence model. A PV power forecasting model, based on the back propagation (BP) artificial neural network (ANN), is proposed in Ref. [14] employing irradiation, temperature, humidity, wind speed, and aerosol index (AI) data. An improved forecasting accuracy is obtained using this model compared with other conventional methods. In Ref. [15], a solar power forecasting model based on different satellite images and a support vector machine (SVM) training scheme is proposed. The predication accuracy of the proposed SVM-based model outperformed the conventional time-series model and an artificial neural network (ANN). Ref. [16] presents a solar power prediction algorithm utilizing spatial and temporal time series meteorological forecast data of three different variables, namely, irradiance on the tilted solar plane, ambient temperature, and wind speed in different locations in Italy. Compared with other benchmark algorithms, significantly improved solar power forecasts are obtained by the proposed algorithm. Ref. [17] establishes improved hybrid model, GA-SVM, combining genetic algorithm (GA) and support SVM to predict the PV system short-term power output, by taking power data of the similar day and the weather data of the

forecasting day as inputs of the model.

References [18,19] proposed forecasting models based on heuristic methods with input data decomposing techniques. Hilbert–Huang Transform (HHT) based neural network approach for forecasting of non-stationary time series has been presented in Ref. [18]. Ref. [19] presents PV solar power forecasting model for PV solar generation system using wavelet decomposition and bias compensated random forest algorithm. The papers discussed forecasting approaches that employ decomposed subseries data of original forecasting input datasets.

The exact comparison of all the forecasting approaches is quite difficult since these approaches depend on different circumstances and the data collection is a tricky task. However, among the literature reported above, it has been noted that the artificial intelligence techniques (ANN, ANFIS, SVM, and etc) outperformed the others in short-term PV solar power prediction. Furthermore, the direct forecasting approaches which are solely based on SCADA measurement of PV power time series data face serious challenges when the historical SCADA record has missed (jumped) data along the training input data set, and unable to provide accurate power forecasts. Similarly, the indirect forecasting methods which entirely rely on meteorological data severely suffer from getting accurate forecasts of the weather variables at the desired location, and unable to offer accurate power predictions. Thus, regarding input data sources for solar power forecasting, techniques which employ both historical SCADA records and meteorological forecasts are more effective and give an improved accuracy as it will be demonstrated in this paper. Considering the algorithms implemented for solar power forecasting, models which utilize combinations of different relevant algorithms are more effective and result in an improved accuracy as it will also be verified in this paper.

In this paper, a novel and robust hybrid short-term PV solar power forecasting strategy based on the combination of wavelet transform (WT) particle swarm optimization (PSO) and support vector machine (SVM) is proposed. The proposed HWPS (hybrid WT-PSO-SVM) forecasting strategy is compared with BPNN (back propagation neural network) [6,7] [14], HGNN (hybrid GA-NN) [20], HPNN (hybrid PSO-NN) [21], SVM [9] [11], HGS (hybrid GA-SVM) [17], HPS (hybrid PSO-SVM) [22], and HHPS (hybrid Hilbert-Huang Transform (HHT)-PSO-SVM) [18] [21] [23], strategies, to show its effectiveness with respect to forecasting accuracy and computation time.

The key contributions of this paper are summarized as follows.

- 1) To propose a novel and robust hybrid strategy for short-term photovoltaic solar power forecasting considering both statistical (historical PV power SCADA records) and physical (historical and future NWP meteorological variables: solar radiation intensity, temperature, cloud cover, humidity, pressure and wind speed);
- 2) To improve PV power forecasting accuracy, considering the forecasting accuracy levels obtained with other seven strategies;
- 3) To provide a practical solution to PV power forecasting challenges for PV power systems having missed SCADA records of output power.

The paper is organized as follows. Section 2 describes the proposed forecasting strategy. Section 3 provides the description and preprocessing of the input data of the forecasting model. The WT-PSO-SVM architecture as a forecasting system and working principles of WT, PSO and SVM are presented in section 4. Sections 5 provides different metrics employed to evaluate the forecast accuracy. The numerical findings and forecasting results for the examined real case-study PV power plant are provided in sections 6. The paper conclusions are outlined in section 7.

2. Proposed forecasting strategy

The proposed PV solar power forecasting strategy is based on the combination of wavelet transform (WT), particle swarm optimization (PSO) and support vector machine (SVM). The WT is utilized to decompose the NWP meteorological variables and SCADA solar power series into a set of improved-characteristic subseries. The decomposed historical values of these subseries are used to train the SVM. The future subseries values of NWP meteorological variable are then used to forecast the future solar power subseries using SVM. The PSO is used to optimize the parameters of the SVM to achieve a higher forecasting accuracy and improved performance. Finally, the forecasted future solar power series is reconstructed by applying the inverse WT.

The general scheme of the proposed hybrid WT-PSO-SVM PV solar power forecasting model is shown in Fig. 1.

Historical NWP meteorological variables (solar radiation intensity, atmospheric temperature, cloud cover, humidity, atmospheric pressure and wind speed) as inputs and historical actual PV power SCADA measurement records as target output are used to train the SVM. Then, the future (forecasted) NWP meteorological variables are applied to the developed (trained) WT-PSO-SVM model in order to forecast the future output power of the PV system.

As aforementioned, the PV solar power generation depends on uncertain environmental conditions and one cannot exactly determine how much power the PV system can generate at a specific time in future. Thus, *concept drift* usually exists in solar power generation systems, that is, the statistical properties of the target variable (PV power), which the proposed model is trying to forecast, change over time in unforeseen ways. This causes big problems unless proper methods are embedded to the proposed model, since the forecasts become less accurate as time passes and the prediction accuracy deteriorates due to this concept drift. To prevent decline in forecast accuracy due to concept drift, the proposed model is continuously updated every 24 h. The model has the capability to be retrained or refreshed periodically on the most recently observed meteorological and SCADA information.

3. Data description and preprocessing

3.1. SCADA system

SCADA as the main data acquisition and control system of PV systems is a critical component and plays a key role for PV solar power forecasting systems. Operating a real-time SCADA system offers the operator access to supervise the PV system by managing

all the system parameters online. This opportunity, for the real PV system studied in this paper, is presented for operator to set appropriate actions in critical situations by a 10 min and/or 1 h record of the PV system information. In addition, this SCADA management system provides comprehensive records of the PV system inverters' status and power outputs as well as PV systems operational availability that acts as a basis for short-term solar power forecast.

In this paper, the SCADA PV power data is a vector containing the PV power historical SCADA records for one year with a time-step of 10 min (52,560 measured power values). The PV system SCADA power record is based on Beijing time zone (BJT), the case study PV system is located in Beijing, China.

3.2. NWP model and meteorological variables

Meteorological data has a considerable impact on solar power prediction. There are a number of methods used to get the meteorological data: observation/measurement, data mining and NWP models. The most fit and reliable technique to get meteorological data is through measurement at the desired site. But, this is not always feasible and even the data is not always accessible.

Data mining technique is relatively flexible, yet its capability to downscale the meteorological data is inadequate. The NWP meteorological models employ equations of physical conservation of energy and this allows a more reasonable downscaling of the weather data. NWP models with higher resolution or smaller radius spherical forecasting zone plays the critical role for the solar power forecasting accuracy improvement.

In recent years, regarding availability of sophisticated computational systems, many solar power forecasting researches are carried out utilizing weather data from NWP models. These research studies used various NWP models like COSMO, WRF, RAMS and MM5. For instance [11–17], present weather-based approaches for PV system output power forecasting utilizing data from NWP meteorological models.

Moreover, several extrapolation methods have been presented by researchers to offer accurate meteorological information at the desired elevation of the PV system using weather data that are gathered at 10 m above the ground surface [24]. In this study, the forecasting system used meteorological predictions of a specific NWP model, called WRF model, obtained in the vicinity of the PV system installation site within 5 km spherical radius resolution.

In this paper, the NWP meteorological data is matrix with a set of values of meteorological variables, arranged in 6 columns. Each column of the matrix is associated with one meteorological variable and contains the variable's historical values for one year with a time-step of 15 min (35,040 historical values). Columns 1 to 6 represent the solar radiation intensity, temperature, cloud cover, humidity, pressure and wind speed values, respectively. Totally, this matrix has 210,240 historical values of weather variables. The NWP (WRF) meteorological data record is based on the Greenwich Mean Time (GMT), 8 h delayed from the BJT.

3.3. Data preprocessing

In the data preprocessing stage, the input data is processed to further simplified forms before the WT decomposition and SVM training stages are triggered:

- 1) Both the SCADA PV power and NWP meteorological data are converted to hourly-average data with 1 h time-step to match the time interval of the two data sources;

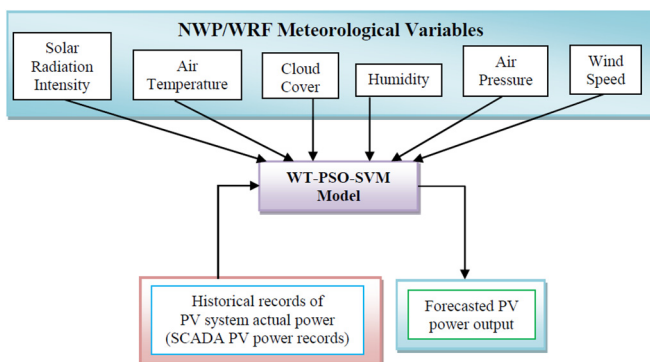


Fig. 1. General scheme of the hybrid WT-PSO-SVM PV solar power forecasting model.

$$x_{hourly}(t) = \frac{\sum_{t=1}^6 x_{10-\min}(t)}{6}, \text{ for the SCADA data} \quad (1)$$

$$x_{hourly}(t) = \frac{\sum_{t=1}^4 x_{15-\min}(t)}{4}, \text{ for the NWP data} \quad (2)$$

where, $x_{10-\min}(t)$, $x_{15-\min}(t)$ and $x_{hourly}(t)$ are the 10 min interval SCADA power data value, the 15 min interval NWP meteorological data value and the corresponding hourly average data value, respectively.

2) The NWP meteorological data records are shifted to the BJT to match with the time zone of the SCADA power data record;

$$x_{BJT}(t) = x_{GMT}(t + 8) \quad (3)$$

where, $x_{BJT}(t)$ and $x_{GMT}(t)$ are the hourly average values of the meteorological data in BJT and GMT, respectively.

3) Missed or jumped data of either source is substituted by an equivalent data using the following jumped data filling technique.

$$x_i = \lambda_1 x_{i-24} + \lambda_2 x_{i+24} \quad (4)$$

where x_i represents the data value of the i th sampling point, λ_1 and λ_2 are the weights for calculation (taken as 0.5 in this paper). x_{i-24} and x_{i+24} represent the data value at the same time on the previous day and the next day, respectively. This shows the capability of the algorithm in the proposed methodology to deal with some flawed data during the training process. It substitutes the flawed data based on the presented technique (4) instantly and uses the substituted data for the SVM training task.

4. Proposed architecture for WT-PSO-SVM

4.1. Wavelet transform (WT)

The WT decompose the NWP meteorological variables series and SCADA PV power series into a set of subseries. These subseries offer better performance characteristics than the original NWP meteorological variables series and SCADA solar power series, and hence, they can be used to forecast the PV power more accurately. The rationale for the better performance characteristics of the subseries is the filtering ability of the WT.

WTs can be classified in two groups: continuous wavelet transform (CWT) and discrete wavelet transform (DWT). The CWT $W(a, b)$ of a signal $f(x)$ with respect to a mother wavelet $\psi(x)$ is defined as follows [25]:

$$W(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} f(x) \psi\left(\frac{x-b}{a}\right) dx \quad (5)$$

where, the scale parameter a manages the distribution of the wavelet and the translation parameter b decides the central position of the wavelet. The DWT $W(m, n)$ is as accurate as the CWT but more efficient [26], and given by:

$$W(m, n) = 2^{-(m/2)} \sum_{t=0}^{T-1} f(t) \psi\left(\frac{t - n2^m}{2^m}\right) \quad (6)$$

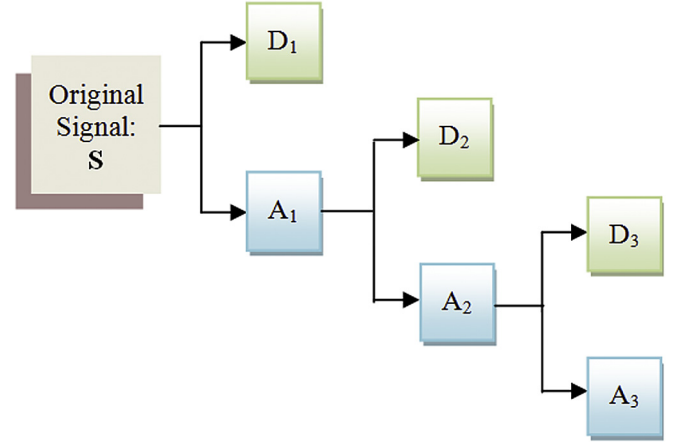


Fig. 2. Wavelet multistage decomposition process.

where, T is the dimension of the signal $f(t)$. The scaling and translation parameters are in terms of the integer parameters m and n (i.e., $a = 2^m$, $b = n2^m$); t is the discrete time sampling index.

A fast decomposition and reconstruction DWT algorithm depending on the four basic filters (decomposition low-pass, decomposition high-pass, reconstruction low-pass, and reconstruction high-pass filters) was implemented by Mallat [27]. Multi-resolution through Mallat's decomposing and reconstructing algorithm is a process to get “approximations” and “details” from a specified signal. By consecutive decomposition of the approximations (Fig. 2), a multistage hierarchical decomposition procedure can be attained where the original signal is decomposed into smaller resolution components.

A Daubechies type wavelet function of order 4 (Db4) is utilized as the mother wavelet $\psi(t)$. This wavelet provides a suitable trade-off between smoothness and wavelength, giving a suitable characteristics for short-term solar power prediction. These types of wavelets have been used by previous research studies for load demand forecasting [25] [26], and electricity price forecasting [28]. Besides, three decomposition levels are used, as in Ref. [28], as it can describe the NWP meteorological variables and SCADA PV power series in a systematic and reasonable way.

4.2. Particle swarm optimization (PSO)

PSO has a small number of parameters to manipulate and is simple to put into implementation. Many research studies and practical applications have been effectively implemented utilizing the PSO concept. The PSO algorithm is simple for mathematical analysis and easy for implementation among other modern optimization methods. It has only two parameters, particles' *position* and *velocity*. Moreover, it is fast convergent and global solution solver for most scientific and engineering optimization problems. That is why it is chosen, in this paper, as the optimization method for searching the optimal parameters of the SVM parameters. Ref. [29] describes a broad outline of PSO and its applications in power systems, and also presents comparative descriptions with other optimization techniques.

For an optimization decision vector $x \in \mathbb{R}^n$ comprising the objective function's optimization decision variables (positions), the feasible search space is defined by a subset below:

$$\Omega = [x_1^L, x_1^U] \times [x_2^L, x_2^U] \times \dots \times [x_n^L, x_n^U] \subset \mathbb{R}^n \quad (7)$$

where, x_j^L and x_j^U are, respectively, the lower and upper bounds of

the search space along the length of dimension j , for $j = 1, 2, \dots, n$.

The i th particle's position or coordinate in the k th iteration is give by:

$$x_i(k) = x_i(k-1) + v_i(k); \quad i = 1, 2, \dots, N \quad (8)$$

where, N is the swarm size and $v_i(k)$ is the i th particle's velocity at the k th iteration which indicates the rate of change of particle's position in the search space, and defined as:

$$v_i(k) = \omega(k)v_i(k-1) + c_1r_1(P_{best,i} - x_i(k-1)) + c_2r_2(G_{best} - x_i(k-1)) \quad (9)$$

where, $\omega(k)$ is a dynamic inertia weight whose value decays linearly with the iteration value to smoothen the velocities over iterations, allowing the swarm to converge more efficiently and accurately, and define as:

$$\omega(k) = \omega_{\max} - \left(\frac{\omega_{\max} - \omega_{\min}}{k_{\max}} \right) \cdot k \quad (10)$$

where, $\omega_{\max}$ is the maximum value of inertia weight; $\omega_{\min}$ is the minimum value of inertia weight; $k_{\max}$ is the maximum number of iterations; c_1 is the cognitive learning rate; c_2 is the social learning rate; r_1 and r_2 are random numbers in the range of 0 and 1; $P_{best,i}$ is the best position achieved by the i th particle; G_{best} is the global best position of all the particles in the swarm.

4.3. Support vector machine (SVM)

SVMs belong to the category of nonparametric techniques, which basically rely on kernel functions. The basics of SVMs have been built up by Vapnik in 1995 [30] and are gaining considerable recognitions recently due to several eye-catching features, and promising practical performances. They have been effectively applied to the problems of forecasting and pattern classification, specifically the classification of two dissimilar classes of patterns. The SVM formulation includes the structural risk minimization (SRM) principle that has been demonstrated to be better to conventional empirical risk minimization (ERM) principle, utilized by conventional neural networks [31]. The fundamental concept of the SVM is to map dataset x into a higher-dimensional characteristic space (hyperplane) through a nonlinear approximation or mapping. The linear regression (classification) in the high-dimensional hyperplane corresponds to the nonlinear regression (classification) in the low-dimensional space, and formulated as follows [32].

$$y(x) = w \cdot \Phi(x) + b \quad (\Phi : R^n \rightarrow R^N) \quad (11)$$

where, $y \in R^N$ is the output training vector; $x \in R^n$ is the input training vector; b is a constant bias term; $w \in R^N$ is the coefficient or weight vector; $\Phi(x)$ is the nonlinear mapping function; and $\Phi : R^n \rightarrow R^N$ is the nonlinear mapping, which transforms the original input training samples to a much higher dimension feature space, hyperspace.

Fig. 3 shows the architectural structure of the SVM, where the regression input x is mapped through the mapping function $\Phi(\cdot)$; the regression output y is the linear combination of weighted $\Phi(x)$.

A particular SVM called *linear epsilon-insensitive SVM* (ϵ -SVM) is employed in this paper because of its scarcity characterization capability. The objective function of the ϵ -SVM is formulated by relying on the ϵ -insensitive loss function. The optimal values of the SVM parameters w and b can be found by solving the minimization problem of the following constrained objective function:

$$\begin{aligned} \min & \left\{ \frac{1}{2} w^T w + \gamma \sum_{i=1}^N (\xi_i + \xi_i^*) \right\} \\ \text{s.t.} & \begin{cases} y_i - w \cdot \Phi(x_i) - b \leq \epsilon + \xi_i \\ w \cdot \Phi(x_i) + b - y_i \leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \end{aligned} \quad (12)$$

where, ξ_i and ξ_i^* are the slack variables; γ is the regularization parameter; N is the training sample size; and ϵ is the loss parameter. Such an optimization problem belongs to the class of quadratic programming, and normally it is solved by solving its counterpart dual problem given as follows:

$$\begin{aligned} \min & \left\{ \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (\alpha_i - \alpha_i^*) \Phi(x_i, x_j) (\alpha_j - \alpha_j^*) \right\} \\ \text{s.t.} & \begin{cases} - \sum_{i=1}^N (\alpha_i + \alpha_i^*) \cdot \epsilon + \sum_{i=1}^N (\alpha_i - \alpha_i^*) \cdot y_i \\ \sum_{i=1}^N (\alpha_i - \alpha_i^*) = 0, \alpha_i, \alpha_i^* \geq 0 \end{cases} \end{aligned} \quad (13)$$

Solving for the nonnegative Lagrange multipliers or coefficients $(\alpha_i - \alpha_i^*)$, the ultimate expression for the SVM estimation of y is given as follows:

$$\hat{y}(x) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) K(x, x_i) + b \quad (14)$$

where, $K(x_i, x_j) = \Phi(x_i) \cdot \Phi(x_j)$ is the SVM kernel function.

According to the Karush-Kuhn-Tucker (KKT) optimality conditions [32] for quadratic programming problems, only some of the coefficients $(\alpha_i - \alpha_i^*)$ will have nonzero values. The data sample points corresponding to the nonzero coefficients with estimation errors equal to or larger than ϵ are called support vectors. The data points in the ϵ -insensitive region are not support vectors and have no contribution to the approximation. Usually, the bigger ϵ , the lesser the number of support vectors and the sparser the depiction of the solutions. For specified n data sample points, the ϵ -SVM

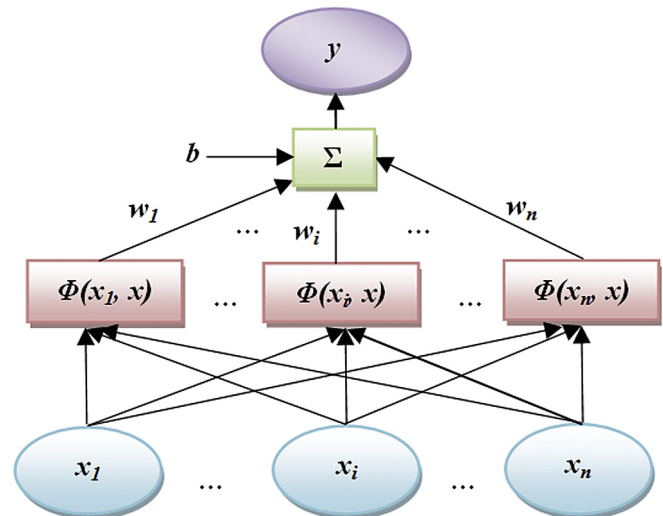


Fig. 3. Architecture of SVM

computes a $2n \times 2n$ kernel matrix. The RBF kernel is employed in this paper, defined as follows. Where, σ is Gauss parameter which is the width of the RBF kernel and determines the dominance region of the support vectors over the data space.

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma^2}\right) \quad (15)$$

In this paper, the ε -SVM model uses the PSO optimization technique to optimally tune its parameters. The parameters of the SVM are formed as parameters of the PSO and the constrained objective function defined in (12) is used as the cost function in PSO. The objective of proposed strategy is to achieve a minimum value for this cost function. This process continues till the forecast error reaches to a desired value. The PSO optimization methods have the advantages of computationally simplicity for a given configuration of SVM structure.

4.4. Proposed hybrid forecasting model

Here, the algorithm of the hybrid model used to realize the proposed solar power forecasting strategy is presented step-by-step in Fig. 4.

As shown in Fig. 4, the wavelet methods are implemented for signal decomposition and reconstruction purposes, respectively, in the first and last stages. Each preprocessed actual time series data of NWP meteorological variables and SCADA PV power are first decomposed independently into several wavelet coefficient signals and one approximation signal.

The original series of the preprocessed NWP meteorological variables (solar radiation intensity, temperature, cloud cover, humidity, pressure and wind speed) and actual SCADA PV power are decomposed separately into four components by the DWT and each subseries is used independently to train the PSO-SVM, shown in Fig. 4. This is implemented to obtain a trained SVM model which enables to forecast power output of the PV system from regional forecast of NWP meteorological variables, shown in Fig. 5.

Then, the subseries of the decomposed future NWP meteorological variables are applied to the developed (trained) PSO-SVM model in order to forecast the future output power subseries of the PV system. Finally, the future power subseries signals are recombined in the last stage to form the final forecasted solar power series of the PV system.

5. Accuracy evaluation metrics

In order to evaluate the accuracy of the proposed HWPS photovoltaic solar power forecasting model, the mean absolute percentage error (MAPE), the sum squared error (SSE), the root mean squared error (RMSE), the standard deviation of error (SDE), and the normalized mean absolute error (NMAE) metrics have been used. These performance evaluation metrics are computed as a function of the actual PV power produced, and defined as follows.

The MAPE metric is defined as:

$$\text{MAPE} = \frac{100}{N} \sum_{h=1}^N \left| \frac{P_h^a - P_h^f}{\bar{P}_h^a} \right| \quad (16)$$

$$\bar{P}_h^a = \frac{1}{N} \sum_{h=1}^N P_h^a \quad (17)$$

where, P_h^a and P_h^f are the actual and forecasted solar power at hour h , respectively, $\bar{P}_h^a$ is the mean actual solar power of the forecast

horizon, and N is the forecast horizon which equals to 24 for daily MAPE.

The SSE metric is defined as:

$$\text{SSE} = \sum_{h=1}^N (P_h^a - P_h^f)^2 \quad (18)$$

The RMSE metric is defined by:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{h=1}^N (P_h^a - P_h^f)^2} \quad (19)$$

The SDE metric is defined by:

$$\text{SDE} = \sqrt{\frac{1}{N} \sum_{h=1}^N (e_h - \bar{e})^2} \quad (20)$$

$$e_h = P_h^a - P_h^f \quad (21)$$

$$\bar{e} = \frac{1}{N} \sum_{h=1}^N e_h \quad (22)$$

where, e_h is the forecast error at hour h and $\bar{e}$ is the mean error of the forecast horizon.

The NMAE metric is defined as:

$$\text{NMAE} = \frac{1}{N} \sum_{h=1}^N \left| \frac{P_h^a - P_h^f}{P_{inst}} \right| \quad (23)$$

where, P_{inst} is the maximum installed power capacity of the PV system.

The variability of a forecasting model, after fitting, is a measure of the uncertainty of the model and can be measured by computing the variance of the forecast error. The forecast is more precise if the value of this variance is smaller [33]. From definition (16), daily error variance can be computed as:

$$\sigma_{e,day}^2 = \frac{1}{N} \sum_{h=1}^N \left(\left| \frac{P_h^a - P_h^f}{\bar{P}_h^a} \right| - (e_{day}) \right)^2 \quad (24)$$

$$e_{day} = \frac{1}{N} \sum_{h=1}^N \left| \frac{P_h^a - P_h^f}{\bar{P}_h^a} \right| \quad (25)$$

6. Case study and numerical results

In this paper, the HWPS model is applied for short-term photovoltaic solar power prediction of a real microgrid PV system in Beijing, China. The PV system contains six PV arrays and has a total generation capacity of 480 kW.

The SVM is trained by time series data of NWP meteorological forecast (solar radiation intensity, temperature, cloud cover, humidity, pressure and wind speed) and actual PV power SCADA measurement, recorded from May 1, 2014 to April 31, 2015.

The forecast information is presented for four days ahead representing the four seasons of a year: July 22, 2015, October 17, 2015, January 5, 2016 and April 14, 2016. Thus, days with specifically best PV power features are not selected intentionally. This provides an uneven accuracy allocation throughout the year that shows the

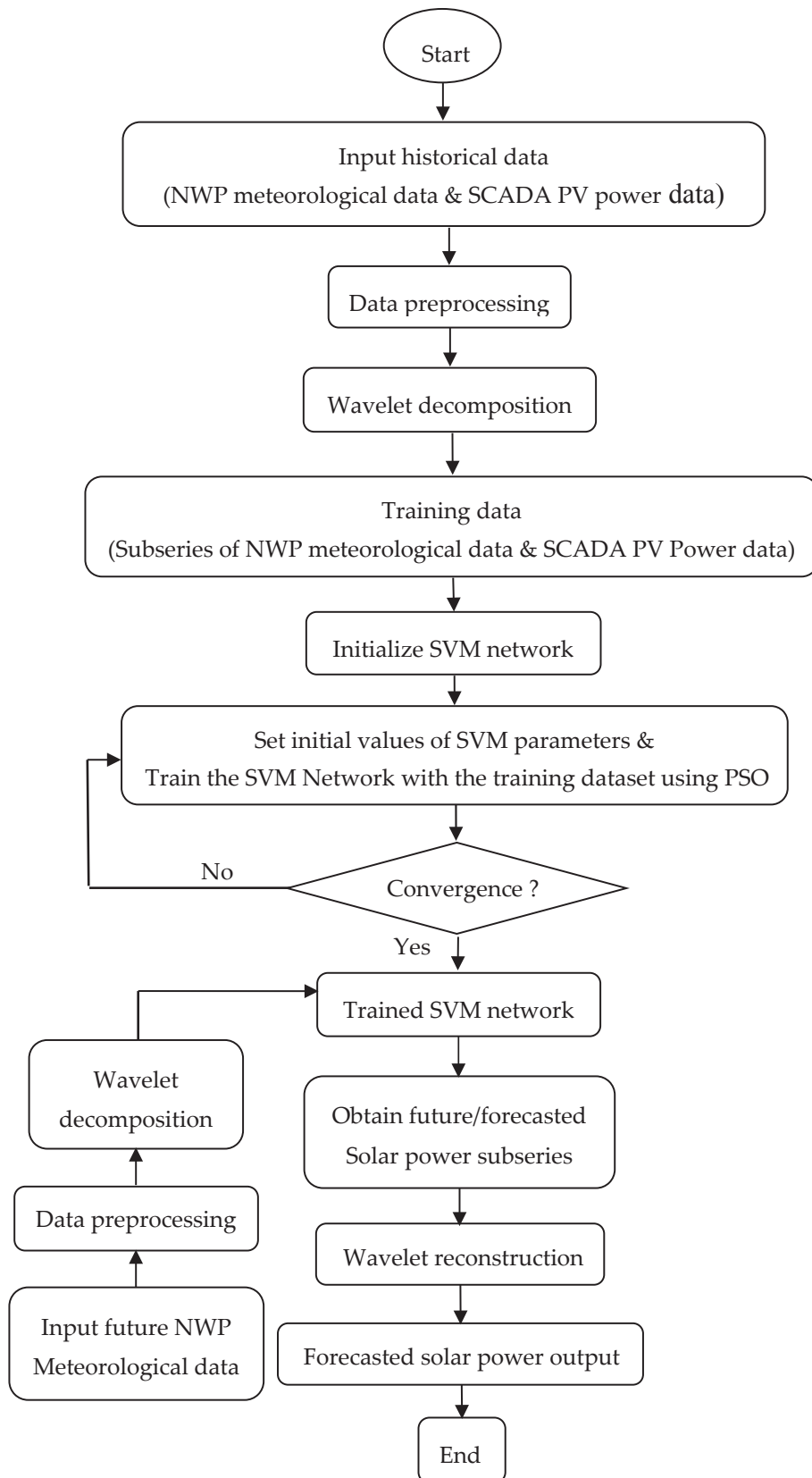


Fig. 4. Implementation procedure of the proposed forecasting strategy.

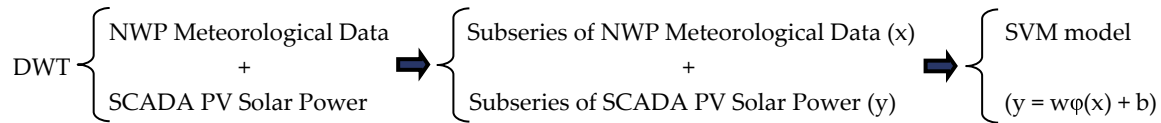


Fig. 5. Illustration of employment of subseries of NWP meteorological data & SCADA PV power data for SVM based PV solar power forecasting model.

reality. The forecasting result is provided for each day with a time-step of 1 h. The numerical values of these testing datasets alongside the corresponding forecasts by the proposed HWPS model are provided in Tables A1–A4 of Appendix A.

The forecasting results by the HWPS model are shown in Figs. 6–9, respectively for the winter, spring, summer and fall days. Each figure illustrates the SCADA actual PV power record vs. the forecasted PV power result by the proposed model.

Table 1 depicts the values of the metrics used to evaluate the accuracy of the proposed HWPS model for forecasting the PV solar power. It gives the statistical error analysis of the forecast results by the proposed model for various case studies, different day types and forecast time horizons. The first column lists the day, the second column shows forecast time horizon, the third column presents the MAPE in percentage, the fourth column presents the square root of the SSE in kW, the fifth column presents the RMSE in kW, and the sixth column presents the SDE in kW.

Interestingly, MAPE, $\sqrt{SSE}$, RMSE and SDE of each case is less than 6%, 16 kW, 4 kW and 4 kW, respectively. It is remarkable that, unlike this paper, considering many research studies accomplished in this field, most of these contributions have been concentrated on specific forecast time horizons or case studies; while, forecast results are not equitably comparable. Results show that proposed model enjoys considerably effective performance for less than 24 h ahead forecasting; whereas, with the increasing forecast time horizon the proposed model performance decreases.

Hence, the proposed short-term photovoltaic solar power forecaster model has an effective performance for forecast durations up to one day (24 h) ahead.

Table 2 gives the performance comparison between the HWPS forecasting strategy and seven other strategies: BPNN, HGNN, HPNN, SVM, HGS, HPS and HHPS, with respect to the daily (24-h ahead) MAPE metric. The proposed strategy results in better forecasting accuracy: the daily MAPE has 4.11% average value. The proposed strategy's average MAPE enhancement with respect to the previous seven strategies is 69.66%, 65.18%, 60.1%, 51.85%, 40.51%, 24.13%, and 8.14%, respectively.

Table 3 shows the forecasting accuracy evaluation by the use of the daily (24-h ahead) NMAE metric, considering the HWPS strategy and the seven other strategies (BPNN, HGNN, HPNN, SVM, HGS, HPS and HHPS).

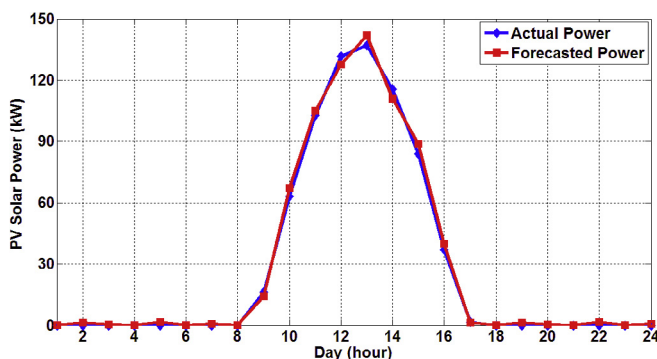


Fig. 6. Actual vs. forecasted PV solar power for a winter day.

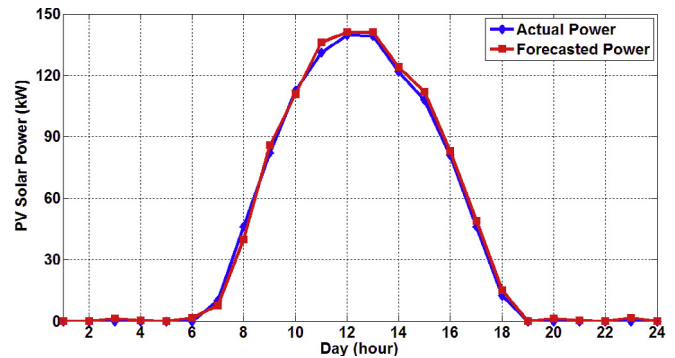


Fig. 7. Actual vs. forecasted PV solar power for a spring day.

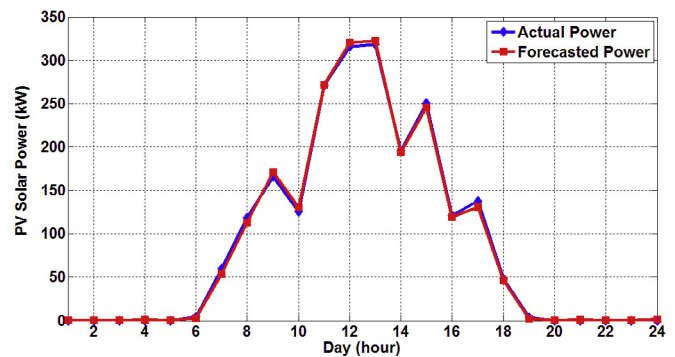


Fig. 8. Actual vs. forecasted PV solar power for a summer day.

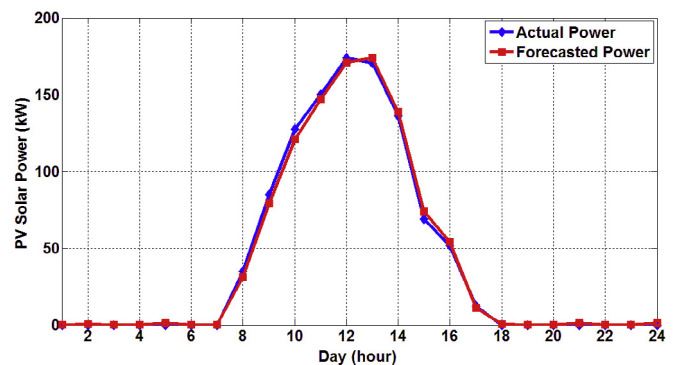


Fig. 9. Actual vs. forecasted PV solar power for a fall day.

Considering the NMAE metric as given in Table 3, the HWPS experienced a daily mean error representing 0.4% of the installed PV power capacity for its 24-h ahead forecasts, over the whole prediction horizon.

In addition to the daily MAPE and NMAE metrics, consistency of prediction results is another important factor to compare forecasting strategies. Table 4 gives a performance comparison between the HWPS strategy and the other seven strategies (BPNN,

Table 1

Statistical error analysis of forecast results by the proposed model for different day types and forecast time horizons.

Day Type	Forecast Time Horizon (hour)	MAPE (%)	$\sqrt{SSE}$ (kW)	RMSE (kW)	SDE (kW)
Winter	3	1.7824	3.6286	0.7392	0.7072
	6	2.4508	4.9894	1.0164	0.9713
	12	3.342	6.8036	1.386	1.326
	24	5.5720	11.3394	2.3146	2.2076
Spring	3	1.3264	3.7630	0.768	0.704
	6	1.8238	5.1742	1.056	0.968
	12	2.4869	7.0557	1.44	1.32
	24	4.1449	11.7595	2.4004	2.2017
Summer	3	0.8701	5.0284	1.024	0.98
	6	1.1964	6.918	1.408	1.204
	12	1.6314	9.426	1.92	1.86
	24	2.7190	15.7137	3.2075	3.1990
Fall	3	1.4208	4.2094	0.8598	0.861
	6	1.9536	5.7904	1.1822	1.1702
	12	2.664	7.896	1.6121	1.5123
	24	4.4419	13.1627	2.6868	2.6818

HGNN, HPNN, SVM, HGS, HPS and HHPS), regarding the daily prediction error variance.

As seen from Table 4, the mean prediction error variance is smaller for the HWPS strategy, demonstrating less uncertainty in the predictions. The proposed strategy's mean error variance enhancement with respect to the previous seven strategies is 87.16%, 82.88%, 77.65%, 69.35%, 55.81%, 32.14% and 13.64%, respectively.

Moreover, as shown in Tables 2–4, the HHT based HHPS forecasting model has given almost similar performance indices with that of the proposed WT based HWPS model. *Hilbert-Huang Transform (HHT)* gives a complete, adaptive and approximately orthogonal subseries representations of the original series of NWP meteorological variables and SCADA PV power. On the other hand, DWT subseries representation of the original series of NWP meteorological variables and SCADA PV power is complete, *a priori* (wavelet is selected first) and orthogonal. HHT is able to determine the instantaneous frequency with high accuracy. However, DWT cannot show the instantaneous frequency, but it can be employed to determine the relative change in the frequency content of the original series. The DWT decomposition is fast, which suits short and very-short term forecasting applications. The HHT decomposition is more accurate, but it is computationally very complex and time consuming. Thus, the WT transform is chosen for the proposed short-term photovoltaic solar power forecasting model due to its computational simplicity compared to the HHT transform for extracting the detailed features of the NWP meteorological data and SCADA PV power series.

The proposed HWPS strategy results in better performance with respect to forecasting accuracy, outperforming the other seven strategies. Moreover, the mean computational time is about 15 s, using MATLAB simulation environment on a personal computer with Intel core i5-5200CPU, 2.20 GHz processor and 4 GB RAM.

Table 2

Comparative daily MAPE results.

Forecasting Strategy	Winter	Spring	Summer	Fall	Average
BPNN	15.3202	16.8117	9.3782	14.1282	13.9096
HGNN	12.9769	14.7015	8.3234	12.4743	12.1190
HPNN	11.3013	12.7168	7.1884	11.0966	10.5758
SVM	9.0845	10.8721	5.8175	9.2860	8.7650
HGS	7.0285	9.0711	4.8269	7.4495	7.0940
HPS	5.8112	7.1868	3.6396	5.6099	5.561,875
HHPS	5.78	5.34	2.73	4.526	4.594
HWPS	5.5720	4.1449	2.7190	4.4419	4.22

Table 3

Comparative daily NMAE (in %) results.

Forecasting Strategy	Winter	Spring	Summer	Fall	Average
BPNN	0.9167	1.5035	1.7382	1.2412	1.3499
HGNN	0.7765	1.3148	1.5427	1.0959	1.1825
HPNN	0.6762	1.1373	1.3323	0.9749	1.0302
SVM	0.5436	0.9723	1.0782	0.8158	0.8525
HGS	0.4206	0.8113	0.8946	0.6545	0.6952
HPS	0.3178	0.6427	0.6746	0.4928	0.5320
HHPS	0.242	0.4721	0.5171	0.411	0.411
HWPS	0.3334	0.3707	0.5039	0.3902	0.40

Table 4

Daily prediction error variance.

Forecasting Strategy	Winter	Spring	Summer	Fall	Average
BPNN	0.0179	0.0204	0.0055	0.0156	0.0148
HGNN	0.0126	0.0153	0.0043	0.0121	0.0111
HPNN	0.0096	0.0116	0.0031	0.0098	0.0085
SVM	0.0065	0.0090	0.0020	0.0072	0.0062
HGS	0.0041	0.0067	0.0014	0.0049	0.0043
HPS	0.0025	0.0047	0.0009	0.0030	0.0028
HHPS	0.0014	0.0023	0.0008	0.0023	0.0022
HWPS	0.0034	0.0014	0.0006	0.0021	0.0019

Hence, the proposed hybrid strategy is both novel and effectively successful for short-term PV solar power forecasting.

7. Conclusions

In this paper, a new hybrid strategy is proposed for short-term photovoltaic solar power forecasting based on the combination of SVM, PSO and WT, considering both statistical (PV power SCADA records) and physical (NWP meteorological variables) data inputs. The implementation of the proposed strategy for short-term solar power prediction is both novel and effectively successful. The daily MAPE and NMAE have 4.22% and 0.4% average values, respectively, outperforming seven other prediction strategies while the mean computation time is smaller than 15 s. Thus, the demonstrated numerical results verify the effectiveness of the proposed strategy for short-term PV solar power prediction.

Conflicts of interest

The authors declare no conflict of interest.

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Appendix A. Forecast model testing dataset (source: Goldwind Microgrid PV solar system)

Table A1

Forecast model testing dataset for a winter day (January 5, 2016)

Hour of the Day	PV Solar Power (kW)	
	Actual	HWPS Forecast
12am	0	0.25
1am	0	1.24
2am	0	0.3
3am	0	0.268
4am	0	1.489
5am	0	0.124
6am	0	0.592
7am	2.45	1.114
8am	16.136	14.35
9am	63.24	67.16
10am	102.76	105
11am	131.832	127.56
12pm	137.104	142
1pm	115.592	111
2pm	84.048	89
3pm	37.096	40
4pm	1.496	1.29
5pm	0.81	0.25
6pm	0.12	1.24
7pm	0	0.3
8pm	0	0.268
9pm	0	1.489
10pm	0	0.124
11pm	0	0.592

Table A2

Forecast model testing dataset for a spring day (April 14, 2016)

Hour of the Day	PV Solar Power (kW)	
	Actual	HWPS Forecast
12am	0	0.114
1am	0	0.25
2am	0	1.24
3am	0	0.3
4am	0	0.268
5am	0	1.489
6am	9.976	7.9
7am	46.24	40
8am	82.056	86
9am	112.56	111
10am	130.96	136
11am	139.712	141
12pm	139.352	141
1pm	121.672	124
2pm	108.04	112
3pm	81.128	83
4pm	46.208	49
5pm	12.376	15
6pm	3.23	2.25
7pm	2.589	1.24
8pm	0	0.3
9pm	0	0.268
10pm	0	1.489
11pm	0	0.124

Table A3

Forecast model testing dataset for a summer day (July 22, 2015)

Hour of the Day	PV Solar Power (kW)	
	Actual	HWPS Forecast
12am	0	0.592
1am	0	0.114
2am	0	0.25
3am	0	1.24
4am	0	0.3
5am	4.232	3
6am	59.808	54
7am	118.568	113
8am	165.448	171
9am	125.2	130
10am	270.976	272
11am	315.144	320
12pm	318.048	322
1pm	196.184	194
2pm	250.656	246
3pm	121.36	119
4pm	137.464	131
5pm	48.224	46
6pm	3.832	2.5
7pm	0	0.25
8pm	0	1.24
9pm	0	0.3
10pm	0	0.268
11pm	0	1.489

Table A4

Forecast model testing dataset for a fall day (October 17, 2015)

Hour of the Day	PV Solar Power (kW)	
	Actual	HWPS Forecast
12am	0	0.124
1am	0	0.592
2am	0	0.114
3am	0	0.25
4am	0	1.24
5am	0	0.3
6am	3.21	1.28
7am	34.848	31
8am	84.904	79
9am	127.696	121
10am	150.384	147
11am	174.12	171
12pm	170.456	174
1pm	136.184	139
2pm	69.128	74
3pm	51.848	54
4pm	12.496	11
5pm	5.23	5.592
6pm	2.32	1.174
7pm	0	0.25
8pm	0	1.24
9pm	0	0.3
10pm	0	0.268
11pm	0	1.489

References

- [1] W.-J. Lee, Y. Liu, Y. Yang, P. Wang, Forecasting power output of PV system based on weather classification and support vector machine, in: Industry applications Society Annual Meeting (IAS), 2011, pp. 1–6.
- [2] Lingzhi Kong, X. Tang, Z. Qi, Study on modified EMAP model and its application in collaborative operation of hybrid distributed power generation system, in: International Conference on Sustainable Power Generation & Supply IEEE, Nanjing, China, 2009.
- [3] X.Y. Guo, A time-interval based probabilistic production simulation of power system with grid-connected photovoltaic generation, Power Syst. Technol. 37 (6) (Jun. 2013) 1499–1505.

- [4] Zhang, Yan Xia, J. Zhao, Application of recurrent neural networks to generated power forecasting for photovoltaic system, *Power Syst. Prot. Control* 39 (15) (Aug. 2011) 96–101.
- [5] R.J. Bessa, Solar power forecasting in smart grids using distributed information, *Int. J. Electr. Power & Energy Syst.* 72 (Aug. 2015) 16–23.
- [6] Ying Yang, L. Dong, Short-term PV generation system direct power prediction model on wavelet neural network and weather type clustering, in: *International Conference on Intelligent Human-machine Systems and Cybernetics IEEE*, Hangzhou, China, 2013.
- [7] Shou Wang, Xiang, Zhang Na, A model to forecast short-term output power of photovoltaic system based on grey and neural theory, *Automat. Electr. Power Syst.* 36 (19) (Oct.2012) 37–41.
- [8] Dong Lei, W.P. Zhou, P. Zhang, et al., Short-term output forecast based on dynamic Bayesian network theory, *Proceeding CSEE* 33 (Jan. 2013) 38–45.
- [9] Matteo De Felice, M. Petitta, P.M. Ruti, Short term predictability of photovoltaic production over Italy, *Renew. Energy* 80 (Sep. 2015) 197–204.
- [10] Marcelo Pinho Almeida, Oscar Perpinan, Luis Narvarte, PV power forecast using a nonparametric PV model, *Sol. Energy* 115 (2015) 354–368.
- [11] Jianwu Zeng, Wei Qiao, Short-term solar power prediction using a support vector machine, *Renew. Energy* 52 (April 2013) 118–127.
- [12] Hong-Tzer Yang, Chao-Ming Huang, Yann-Chang Huang, Yi-Shiang Pai, A weather-based hybrid method for 1-day ahead hourly forecasting of PV power output, *IEEE Trans. Sustain. energy* 5 (3) (July 2014).
- [13] Chen Yang, Anupam A. Thatte, Le Xie, Multitime-scale data-driven spatio-temporal forecast of photovoltaic generation, *IEEE Trans. Sustain. energy* 6 (1) (January 2015).
- [14] Jun Liu, Wanliang Fang, Xudong Zhang, Chunxiang Yang, An improved photovoltaic power forecasting model with the assistance of aerosol index data, *IEEE Trans. Sustain. energy* 6 (2) (April 2015).
- [15] Han Seung Jang, Kuk Yeol Bae, Hong-Shik Park, Dan Keun Sung, Solar power prediction based on satellite images and support vector machine, *IEEE Trans. Sustain. energy* 7 (3) (July 2016).
- [16] Akin Tascikaraoglu, Borhan M. Sanandaji, Gianfranco Chicco, Valeria Cocina, Filippo Spertino, Ozan Erdinc, Nikolaos G. Paterakis, João P.S. Catalão, Compressive spatio-temporal forecasting of meteorological quantities and photovoltaic power, *IEEE Trans. Sustain. energy* 7 (3) (July 2016).
- [17] Jidong Wang, Ran Ran, Zhilin Song, Jiawen Sun, "Short-Term photovoltaic power generation forecasting based on environmental factors and GA-SVM, *J. Electr. Eng. Technol.* 12 (01) (January 2017) 64–71.
- [18] V.G. Kurbatskii, D.N. Sidorov, V.A. Spiryayev, N.V. Tomin, On the neural network approach for forecasting of nonstationary time series on the basis of the hilbert–huang transform, *Automat. Remote Control* 72 (7) (2011) 1405–1414.
- [19] P.H. Chiang, S.P.V. Chiluvuri, S. Dey, T.Q. Nguyen, Forecasting of solar photovoltaic system power generation using wavelet decomposition and bias compensated random forest, in: *2017 IEEE GreenTech*, March 29–31, 2017, pp. 260–266. Denver, USA.
- [20] A.T. Eseye, J. Zhang, D. Zheng, H. Ma, G. Jingfu, Short-term wind power forecasting using a double-stage hierarchical hybrid GA-ANN approach, in: *2017 IEEE 2nd International Conference on Big Data Analysis (ICBDA 2017)*, Beijing, China, March 10–12, 2017.
- [21] A.T. Eseye, J. Zhang, D. Zheng, H. Li, G. Jingfu, A double-stage hierarchical hybrid PSO-ANN model for short-term wind power prediction, in: *2017 the 2nd IEEE International Conference on Cloud Computing and Big Data Analysis (ICCCBDA 2017)*, Chengdu, China, April 28–30, 2017.
- [22] X. Zhang, Y. Guo, Optimization of SVM parameters based on PSO algorithm, in: *Fifth International Conference on Natural Computation*, Tianjin, China, Aug. 2009.
- [23] A. Forough, E. Mohammadi, S. Esmaeili, Application of Hilbert-Huang transform and support vector machine for detection and classification of voltage sag sources, *Turkish J. Electr. Eng. Comput. Sci.* 22 (Aug. 2014) 1116–1129.
- [24] Y. Charabi, Arabian summer monsoon variability: teleconexion to ENSO and IOD, *Atmos. Res.* (2009) 105–117.
- [25] N. Amjady, F. Keynia, Short-term load forecasting of power systems by combination of wavelet transform and neuro-evolutionary algorithm, *Energy* 34 (1) (Jan. 2009) 46–57.
- [26] A.J.R. Reis, A.P.A. da Silva, Feature extraction via multiresolution analysis for short-term load forecasting, *IEEE Trans. Power Syst.* 20 (1) (Feb. 2005) 189–198.
- [27] S. Mallat, "A theory for multiresolution signal decomposition—the wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no 7 (Jul. 1989) 674–693.
- [28] N. Amjady, F. Keynia, Day ahead price forecasting of electricity markets by a mixed data model and hybrid forecast method, *Int. J. Electr. Power Energy Syst.* 30 (9) (Nov. 2008) 533–546.
- [29] Y. Del Valle, G.K. Venayagamoorthy, S. Mohagheghi, J.-C. Hernandez, R.G. Harley, Particle swarm optimization: basic concepts, variants and applications in power systems, *IEEE Trans. Evol. Comput.* 12 (2) (Apr. 2008) 171–195.
- [30] V. Vapnik, *The Nature of Statistical Learning Theory*, Springer-Verlag, New York, 1995.
- [31] S.R. Gunn, *Support Vector Machines for Classification and Regression*, Technical Report, University of Southampton (Faculty of Engineering, Science and Mathematics), Southampton, 1998.
- [32] K.R. Miller, V. Vapnik, *Using Support Vector Machine for Time Series Prediction*, MIT, Cambridge, 1999, pp. 243–253.
- [33] H.M.I. Pousinho, V.M.F. Mendes, J.P.S. Catalão, A hybrid PSO–ANFIS approach for short-term wind power prediction in Portugal, *Elsevier Energy Convers. Manag.* 52 (1) (January 2011) 397–402.